Semisupervised Feature Extraction of Hyperspectral Image Using Nonlinear Geodesic Sparse Hypergraphs

Yule Duan, Hong Huang, *Member, IEEE*, and Tao Wang

***Abstract*— Recently, the sparse representation (SR)-based graph embedding method has been extensively used in feature extraction (FE) tasks, but it is hard to reveal the complex manifold structure and multivariate relationship of samples in the hyperspectral image (HSI). Meanwhile, the small size sample problem in HSI data also limits the performance of the traditional SR approach. To tackle this problem, this article develops a new semisupervised FE algorithm called a geodesic-based sparse manifold hypergraph (GSMH). The presented method first utilizes the geodesic distance to measure the nonlinear similarity between samples lying on manifold space and further constructs the manifold neighborhood of each sample. Then, a geodesic-based neighborhood SR (GNSR) model is designed to explore the multivariate sparse correlations of different manifold neighborhoods. Considering the multivariate sparse manifold correlations among samples, a pair of semisupervised hypergraphs (HGs) is constructed to effectively incorporate the labeled and unlabeled training information in the embedding process and obtain the nonlinear discriminative feature representation for HSI. Experimental results on three HSI datasets indicate that the proposed method not only achieves satisfying FE performance with limited labeled training samples but also shows superiority compared with other state-of-the-art methods.**

***Index Terms*— Feature extraction (FE), graph theory, hyperspectral image (HIS), manifold learning, sparse representation (SR).**

I. Introduction

BENEFITING from the development of hyperspectral imaging technology, the hyperspectral image (HSI) records many contiguous spectral bands and offers rich spectral features to distinguish the slight difference between various land cover types [1], [2]. However, the highly correlated spectral bands in HSI possess too much redundant information that causes some serious challenges to classification [3], such as the small sample size problem and huge computational burden [4], [5]. Feature extraction (FE) is an efficient solution for the above-mentioned problems in HSI [6], [7]. The aim of FE is to map original HSI data into a low-dimensional space without sacrificing the desirable information [8], [9].

FE methods can be roughly categorized into supervised and unsupervised types, according to whether using *a priori* label information or not [10]. Principle component analysis (PCA) [11], isometric feature mapping (ISOMAP) [12], locality preserving projection (LPP), neighborhood preserving embedding (NPE), and unsupervised FE methods are widely used for HISs [13], [14], but they ignore the label information and cannot enhance the discriminant of features. Unlike unsupervised methods, supervised FE methods utilize the *a priori* information of labeled samples to guide the feature learning process, such as linear discriminant analysis (LDA) [15], marginal fisher analysis (MFA) [16], local fisher discriminant analysis (LFDA) [17], and local geometric structure Fisher analysis (LGSFA) [18]. LDA utilizes scatter matrices to reveal the discriminant characteristic of samples, while LFDA, MFA, and LGSFA construct different graphs to reveal the interclass and intraclass relationships between data points.

However, these mentioned FE algorithms need to choose a suitable similarity measurement strategy and the number of neighbors by virtue of experience, which is difficult to select correct neighbors for graph construction [19]. Sparse representation (SR) is an effective approach to select sparse neighbors adaptively [20], [21]. Sparse graph-based discriminant analysis (SGDA) reveals the neighbor relationship via SR and constructs two graphs to adaptively maintain the sparse similarities between data points [22]. Discriminative learning by SR projections (DLSP) extracts features through preserving the sparse reconstructive relationship of data points [23]. Local manifold-based sparse discriminant learning (LMSDL) conducts local weighted SR for revealing the local sparse relationship samples, and then it constructs a pair of interclass and intraclass geometrical sparse graphs to preserve the discriminant relationship between samples in embedding space [24].

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In real-world HSI applications, the labeling samples task requires considerable human resources and time consumption, which leads to the labeled data are usually insufficient for training FE models [25]. While abundant unlabeled data can be available in a short time at low cost [26], [27], many semisupervised FE algorithms have been proposed for HSI. Semisupervised discriminant analysis (SDA) and semisupervised local discriminant analysis (SELD) as the semisupervised variants of LDA simultaneously utilize labeled and unlabeled samples for training [28], [29]. But these approaches both ignore the important data structure information of HSI. Based on the graph embedding framework, semisupervised LPP (SSLPP) and semisupervised LFDA (SELF) are designed for revealing the manifold structure of labeled and unlabeled data [30], [31]. Geodesic-based manifold joint hypergraphs (GMJHs) measure the similarity between samples by means of geodesic distance, and it further constructs supervised and unsupervised hypergraphs (HGs) for jointly learning embedding features of HSI [32]. Moreover, researchers present sparse coding-based geometric subspace projection (SCGSP) by introducing SR into semisupervised graph learning framework [33], which reveals the discriminant sparse correlation of all samples.

However, the above-mentioned FE semisupervised algorithms mostly utilize a graph framework to represent the one-to-one relationship in data. In real applications, data points generally possess many-to-many relationships, for example, one point belongs to multiple neighborhoods or one neighborhood contains multiple points [34], [35]. To handle this problem, the HG model is proposed to represent the high-order relationship by setting each hyperedge to represent affinity information of a set of data points [36], [37]. Discriminant hyper-Laplacian projection (DHLP) builds an interclass HG for representing the multivariate discriminant relation between points [38]. Semisupervised HG embedding (SHGE) constructs two HGs to reveal the multivariate similarities of labeled and unlabeled samples, respectively [34]. But DHLP and SHGE both ignore the sparse or geometrical structure in HSI. Multistructure manifold embedding (MSME) constructs two HGs to reveal the multiple sparse and manifold structures of unlabeled samples, respectively. But it abandons the valuable label information and fails to fuse the manifold and sparse properties via combining graphs [39]. Local constraint-based sparse manifold HG learning (LCSMHL) introduces local weight distance and local spectral constrain item into sparse optimization to discover the local geometrical and sparse structures [40]. With the obtained sparse coefficients, LCSMHL constructs two manifold-based sparse HGs to learn the embedding feature, but it only focuses on preserving local structure instead of global nonlinear structure. Semisupervised sparse manifold discriminative analysis (S^3MDA) constructs a pair of supervised and unsupervised graphs based on the sparse coefficients obtained from manifold-based SR (MSR) model [41], [42]. However, the MSR model only enlarges the sparse coefficients between similar points via Euclidean distance weights, thus the produced sparse coefficients cannot indicate the real manifold structure in HSI. Moreover, the S^3MDA algorithm constructs supervised and unsupervised graphs without taking into account the multivariate relationship between labeled and unlabeled samples.

For addressing the aforementioned issues, we put forward a novel semisupervised FE method using geodesic-based sparse manifold hypergraphs (GSMHs). For HSI data lying on manifold space, GSMH first measures the nonlinear similarity of samples via geodesic distance and further constructs the manifold neighborhood of each sample. Then, it designs a geodesic-based neighborhood SR (GNSR) model to reveal the sparse relationship between different manifold neighborhoods. Subsequently, it implies the probabilistic relationship between unlabeled samples and each class via the geodesic-based manifold sparse coefficients. With the probabilistic class relationship, a pair of semisupervised hypergraphs is constructed to simultaneously reveal the multivariate sparse correlation and local discriminant manifold structure in HSI. The key contributions of the proposed algorithm are summarized as follows.

1) A GNSR model is built by fusing geodesic distance weights and local manifold reconstruction. This modified SR model considers the local manifold structure in the sparse optimization process and further discovers the multivariate sparse relationship between two manifold neighborhoods.
2) GSMH implies the probabilistic relationship between unlabeled samples and each class via the geodesic-based manifold sparse coefficients. With the probabilistic class relationship and manifold reconstruction weights, a sparse manifold neighborhood HG of unlabeled samples is constructed to further discover the potential discriminant information and represent the multivariate sparse manifold relation between unlabeled samples.
3) GSMH constructs a discriminant manifold HG according to the nonlinear geodesic similarity of the labeled sample. Relying on the designed HGs of labeled and unlabeled samples, a semisupervised FE model is designed by preserving the multivariate sparse manifold correlation samples, which effectively incorporates the labeled and unlabeled training information in data embedding and brings significant benefits for subsequent HSI classification.

This article is organized as follows: Section II introduces the proposed method GSMH in detail. Section III demonstrates the experimental results of GSMH and other state-of-the-art FE methods. Finally, some conclusions are drawn in Section IV.

II. Geodesic-Based Sparse Manifold Hypergraphs

Based on the SR and manifold learning theories, we design a GSMHs algorithm to discover the multivariate relationship and discriminant embedding feature of high-dimensional HSI data. The detailed flowchart of GSMH is shown in Fig. 1. In this section, an HSI dataset with D spectral bands and $m+n$ samples is defined as $\mathbf{X} = [\mathbf{X}_l, \mathbf{X}_u] \in \Re^{D\times(m+n)}$. $\mathbf{X}$ can be divided into labeled samples set $\mathbf{X}_l = [\mathbf{x}_1^l, \mathbf{x}_2^l, \ldots, \mathbf{x}_m^l]$ and unlabeled samples set $\mathbf{X}_u = [\mathbf{x}_1^u, \mathbf{x}_2^u, \ldots, \mathbf{x}_n^u]$. The class label of $\mathbf{x}_i$ is set as $l(\mathbf{x}_i) \in \{1, 2, \ldots, c\}$, and c denotes the number of classes.

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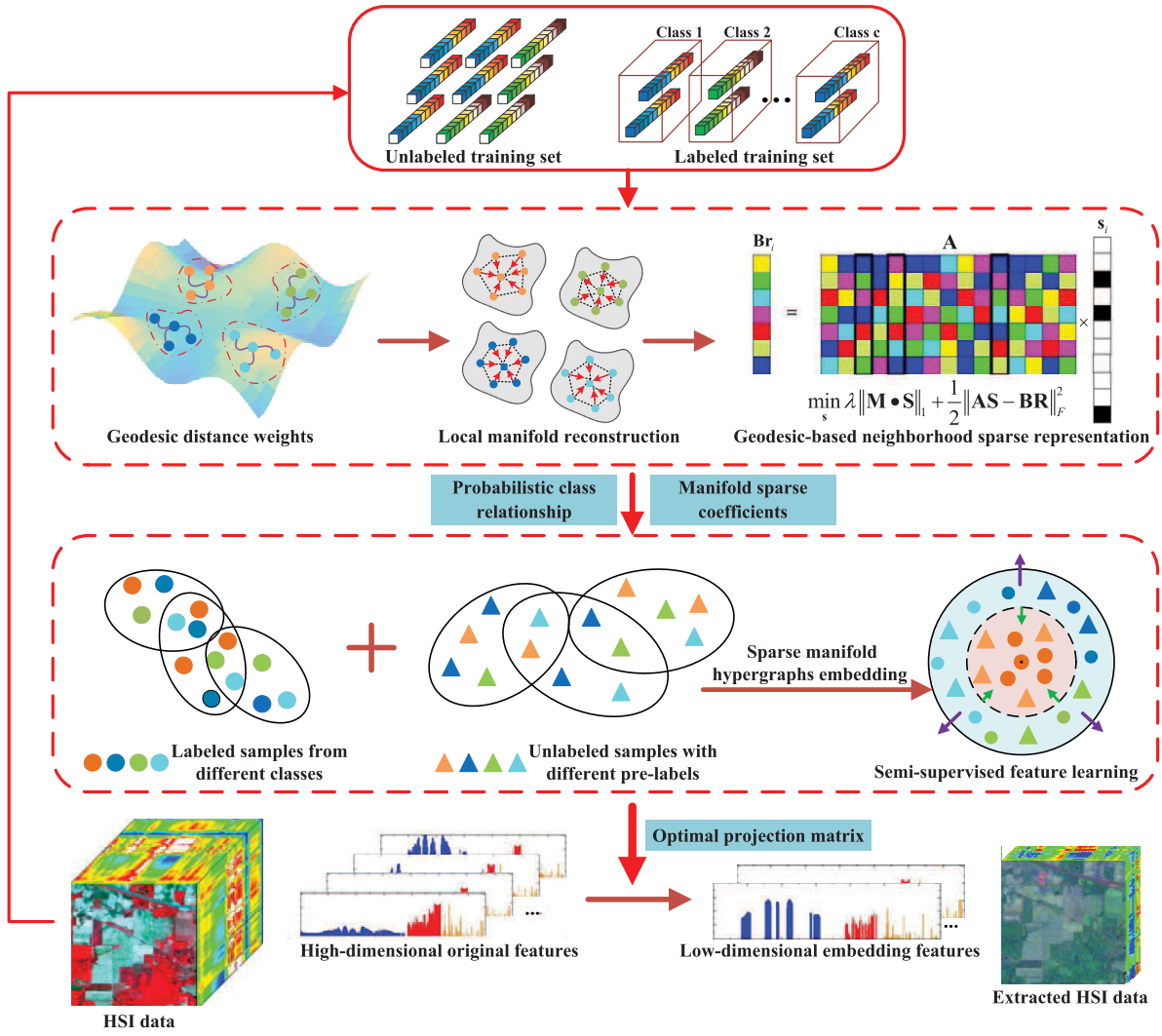


Fig. 1. Flowchart of the proposed method GSMH.

A. Geodesic-Based Neighborhood Sparse Representation

Suffering from the nonlinear spectrum transmitting in the atmosphere, hyperspectral imagery usually contains complex nonlinear manifold structures in the high-dimensional spectral feature space. However, traditional SR models that represent target signals with a linear combination of fewer basic signals are inadequate to consider the manifold structure in HSI. Geodesic distance as a famous nonlinear measurement is able to reveal a similar relation between samples lying on the nonlinear manifold. Therefore, we introduce geodesic distance into sparse representation for designing a new GNSR model, which aims to discover the nonlinear sparse relationship between local manifold neighborhoods.

To calculate the geodesic distance between samples, we first define the neighborhood $N(\mathbf{x}_i)$ of each sample according the k-nearest neighbor (NN) method. The number of neighbors k is set to $k = n/c$, in which n and c, respectively, denote the numbers of unlabeled samples and classes. For all samples in $\mathbf{X}$, the geodesic distance matrix $\mathbf{D}_G = [d_{\text{geo}}(\mathbf{x}_i, \mathbf{x}_j)]_{(m+n)\times(m+n)}$ is defined as

$$d_{\text{geo}}(\mathbf{x}_i, \mathbf{x}_j) = \begin{cases} d(\mathbf{x}_i, \mathbf{x}_j), & \mathbf{x}_j \in N(\mathbf{x}_i) \\ \min\left\{d(\mathbf{x}_i, \mathbf{x}_j), d(\mathbf{x}_i, \mathbf{x}_k) + d(\mathbf{x}_k, \mathbf{x}_j)\right\}_{k=1}^{m+n}, & \mathbf{x}_j \notin N(\mathbf{x}_i) \end{cases} \tag{1}$$

in which $d(\mathbf{x}_i, \mathbf{x}_j)$ represents the Euclidean distance between two samples. The calculation of geodesic distance is essentially an optimization problem of the shortest path between discrete points, which can be solved by the Floyd–Warshall algorithm [43]. The geodesic distance $d_{\text{geo}}(\mathbf{x}_i, \mathbf{x}_j)$ between samples $\mathbf{x}_i$ and $\mathbf{x}_j$ reflects the nonlinear distance between points lying on manifold. Then, a geodesic distance weight matrix $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_n]$ is designed to represent the manifold similarities between unlabeled samples and each labeled sample, while the weight vector $\mathbf{m}_j$ of unlabeled sample $\mathbf{x}_j^u$ is set as

$$\mathbf{m}_j = \left[\frac{d_{\text{geo}}(\mathbf{x}_1^l, \mathbf{x}_j^u)}{\sum_{i=1}^m d_{\text{geo}}(\mathbf{x}_i^l, \mathbf{x}_j^u)}, \ldots, \frac{d_{\text{geo}}(\mathbf{x}_m^l, \mathbf{x}_j^u)}{\sum_{i=1}^m d_{\text{geo}}(\mathbf{x}_i^l, \mathbf{x}_j^u)}\right]^{\mathrm{T}}. \tag{2}$$

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In geodesic distance weight matrix $\mathbf{M}$, two points possessing small geodesic distances on manifold have a small weight that indicates large nonlinear similarity. Through minimizing the l_1-norm of the product of weight matrix $\mathbf{M}$ and sparse coefficient matrix, the SR model enforces the sparse coefficient of closed points having large value, which ensures sparse coefficients can reflect the nonlinear similarities of samples on the manifold.

For considering the local manifold structure during sparse optimal process, we reconstruct unlabeled samples with their local manifold neighborhood and then introduce the local neighborhood information into SR to explore the reconstruction point. For unlabeled sample $\mathbf{x}_j^u$, k nearest geodesic unlabeled neighbors are selected to construct the manifold neighborhood $N(\mathbf{x}_i^u)$. The local reconstruction vector $\mathbf{r}_j = [r_{1j}, r_{2j}, \ldots, r_{nj}]^{\mathrm{T}} \in \Re^{m\times 1}$ of $\mathbf{x}_j^u$ is defined by

$$r_{ij} = \begin{cases} \dfrac{1/d_{\mathrm{geo}}\left(\mathbf{x}_i^u, \mathbf{x}_j^u\right)}{\sum_{\mathbf{x}_t^u \in N\left(\mathbf{x}_j^u\right)} 1/d_{\mathrm{geo}}\left(\mathbf{x}_t^u, \mathbf{x}_j^u\right)}, & \mathbf{x}_i^u \in N\left(\mathbf{x}_j^u\right) \\ 0, & \mathbf{x}_i^u \notin N\left(\mathbf{x}_j^u\right). \end{cases} \tag{3}$$

According to the above-mentioned formula, the neighbor having a smaller geodesic distance between the center sample has a larger reconstruction weight. The reconstruction weights in matrix $\mathbf{R}$ reflect the nonlinear similarity in neighborhoods and further promote similar neighbors to play great roles in reconstruction. Therefore, the reconstruction weight matrix $\mathbf{R}$ ensures the reconstruction point represents the neighborhood manifold structure information reasonably.

With the reconstruction vector and unlabeled dataset, each unlabeled sample $\mathbf{x}_j^u$ can be reconstructed as $\hat{\mathbf{x}}_j^u = \mathbf{X}_u\mathbf{r}_j$. And the reconstruction points of whole unlabeled dataset are denoted by $\hat{\mathbf{X}}_u = \mathbf{X}_u\mathbf{R}$, in which local manifold reconstruction matrix is $\mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_n] \in \Re^{n\times n}$. In the local manifold reconstruction, the nonneighbor points do not participate in reconstruction for avoiding large reconstruction error. Meanwhile, the neighbor point possessing larger geodesic weight will play a larger role in reconstruction. Therefore, the reconstruction points can effectively represent the nonlinear manifold structure of data points in neighborhood.

Based on the above-mentioned weight matrix $\mathbf{M}$ and reconstruction matrix $\mathbf{R}$, the GNSR model is constructed with the following objective function:

$$\begin{cases} \min \lambda\|\mathbf{M} \bullet \mathbf{S}\|_1 + \dfrac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\mathbf{R}\|_F^2 \\ \text{s.t. } \mathbf{i}_m^{\mathrm{T}}\mathbf{Z} = \mathbf{i}_n^{\mathrm{T}}, \quad \mathbf{Z} = \mathbf{S} \end{cases} \tag{4}$$

where $\bullet$ represents the dot product of matrices, $\mathbf{S}$ is the sparse coefficients matrix, $\mathbf{Z}$ is an auxiliary matrix, $\mathbf{i}_n = [1, 1, \ldots, 1]_{n\times 1}^{\mathrm{T}}$ and $\mathbf{i}_m = [1, 1, \ldots, 1]_{m\times 1}^{\mathrm{T}}$ are all-one vectors, and λ is a regularization coefficient.

Fig. 2 gives a visual example to show the sparse optimization process of GNSR. In nonlinear data space, the samples from the same class are usually distributed in the close area of the manifold. The geodesic distance weight matrix $\mathbf{M}$ promotes GNSR to select labeled neighbors that possess stronger sparse correlation and small geodesic distance with the unlabeled sample. Thus, the calculated sparse coefficients can

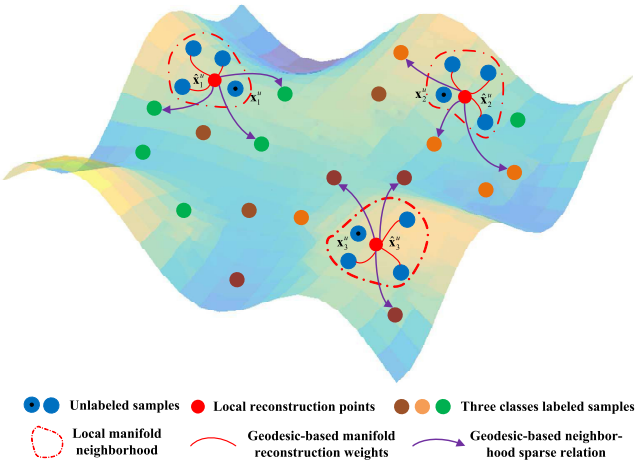


Fig. 2. Example to explain the sparse optimization process of GNSR.

reflect the nonlinear manifold similarities between samples. At the same time, GNSR calculates the SR errors based on the reconstruction sample that represents the combinatorial nature of local manifold neighbors. Differing from the traditional SR model, GNSR discovers the multivariate sparse relationship between labeled samples and unlabeled local manifold neighborhoods, rather than two samples. Therefore, GNSR fully considers the local manifold structure of data and further improves the multivariate correlation representation power of sparse coefficients.

To meet the requirement of convex optimization, we add two penalty terms and transform the objective function as

$$\begin{cases} \min \ \lambda\|\mathbf{M} \bullet \mathbf{S}\|_1 + \dfrac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\mathbf{R}\|_F^2 \\ \qquad\quad + \dfrac{\rho}{2}\left(\left\|\mathbf{i}_m^{\mathrm{T}}\mathbf{Z} - \mathbf{i}_n\right\|_2^2 + \|\mathbf{Z} - \mathbf{S}\|_F^2\right) \\ \text{s.t. } \mathbf{i}_m^{\mathrm{T}}\mathbf{Z} = \mathbf{i}_n, \quad \mathbf{Z} = \mathbf{S}. \end{cases} \tag{5}$$

Then, the alternating direction method of multipliers (ADMMs) approach can be used for calculating optimization solution, and the Lagrangian function of (5) can be written as:

$$\begin{aligned} \mathcal{L}(\mathbf{Z}, \mathbf{S}, \mathbf{T}_1, \mathbf{T}_2) = {} & \lambda\|\mathbf{M} \bullet \mathbf{S}\|_1 + \frac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\mathbf{R}\|_F^2 \\ & + \frac{\rho}{2}\left\|\mathbf{i}_m^{\mathrm{T}}\mathbf{Z} - \mathbf{i}_n^{\mathrm{T}}\right\|_F^2 + \mathrm{tr}\left(\mathbf{T}_1^{\mathrm{T}}\left(\mathbf{i}_m^{\mathrm{T}}\mathbf{Z} - \mathbf{i}_n^{\mathrm{T}}\right)\right) \\ & + \frac{\rho}{2}\|\mathbf{Z} - \mathbf{S}\|_F^2 + \mathrm{tr}\left(\mathbf{T}_2^{\mathrm{T}}(\mathbf{Z} - \mathbf{S})\right). \end{aligned} \tag{6}$$

By some algebraic operations, (6) can be reformulated as

$$\begin{aligned} \mathcal{L}(\mathbf{Z}, \mathbf{S}, \mathbf{T}_1, \mathbf{T}_2) = {} & \lambda\|\mathbf{M} \bullet \mathbf{S}\|_1 + \frac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\mathbf{R}\|_F^2 \\ & + \frac{\rho}{2}\left\|\mathbf{i}_m^{\mathrm{T}}\mathbf{Z} - \mathbf{i}_n^{\mathrm{T}} + \frac{\mathbf{T}_1}{\rho}\right\|_F^2 - \frac{\|\mathbf{T}_1\|_F^2}{\rho^2} \\ & + \frac{\rho}{2}\left\|\mathbf{Z} - \mathbf{S} + \frac{\mathbf{T}_2}{\rho}\right\|_F^2 - \frac{\|\mathbf{T}_2\|_F^2}{\rho^2}. \end{aligned} \tag{7}$$

In each iteration, ADMM updates one variable by minimizing the Lagrangian function $\mathcal{L}$ with other variables fixed.

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At first, we obtain $\mathbf{Z}^{k+1}$ by

$$\mathbf{Z}^{k+1} = \arg\min \frac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\mathbf{R}\|_F^2 + \frac{\rho}{2}\left\|\mathbf{i}_m^{\mathrm{T}}\mathbf{Z} - \mathbf{i}_n^{\mathrm{T}} + \frac{\mathbf{T}_1}{\rho}\right\|_F^2 + \frac{\rho}{2}\left\|\mathbf{Z} - \mathbf{S} + \frac{\mathbf{T}_2}{\rho}\right\|_F^2. \quad (8)$$

By setting the partial derivative of $\mathcal{L}$ at $\mathbf{Z}^{k+1}$ as zero, $\mathbf{Z}^{k+1}$ can be computed by

$$\mathbf{Z}^{k+1} = \left(\mathbf{X}_l^{\mathrm{T}}\mathbf{X}_l + \rho\mathbf{i}_m\mathbf{i}_m^{\mathrm{T}} + I\right)^{-1}\left(\mathbf{X}_l^{\mathrm{T}}\mathbf{X}_u\mathbf{R} + \rho\left(\mathbf{i}_m\mathbf{i}_n^{\mathrm{T}} + \mathbf{S}^k\right) - \left(\mathbf{i}_m\mathbf{T}_1^{kT} + \mathbf{T}_2^k\right)\right). \quad (9)$$

For $\mathbf{S}^{k+1}$, its kth iteration formulation is obtained as

$$\mathbf{S}^{k+1} = \lambda\|\mathbf{M} \bullet \mathbf{S}^k\|_1 + \frac{\rho}{2}\left\|\mathbf{Z}^{k+1} - \mathbf{S}^k + \frac{\mathbf{T}_2^k}{\rho}\right\|_F^2 = S_{\frac{\lambda\mathbf{M}}{\rho}}\left(\mathbf{Z}^{k+1} + \frac{\mathbf{T}_2^k}{\rho}\right) \quad (10)$$

where $\mathcal{S}$ is the soft-thresholding operator. Finally, $\mathbf{T}_1$ and $\mathbf{T}_2$ are iteratively updated as follows:

$$\mathbf{T}_1^{k+1} = \mathbf{T}_1^k + \rho\left(\mathbf{i}_m^{\mathrm{T}}\mathbf{Z}^{k+1} - \mathbf{i}_n^{\mathrm{T}}\right) \quad (11)$$

$$\mathbf{T}_2^{k+1} = \mathbf{T}_2^k + \rho\left(\mathbf{Z}^{k+1} - \mathbf{S}^{k+1}\right). \quad (12)$$

The above-mentioned iteration steps are repeated until the convergence is achieved or iterative number reaches maximum value. The complete procedure of solving GNSR with ADMM is shown in Algorithm 1.

Algorithm 1 Solving GNSR With ADMM

1: **Initialize:** set $\mathbf{Z}^0$, $\mathbf{S}^0$, $\mathbf{T}_1^0$, $\mathbf{T}_2^0$to be zero, $k_{\max} = 10^3$, $k = 0$, $\rho = 1.1$, $\varepsilon = 10^{-3}$ and Terminate=False,.
2: **While** (Terminate = False) do
3: Update $\mathbf{Z}^{k+1}$ with Eqs. (8)-(9)
4: Update $\mathbf{S}^{k+1} = \mathcal{S}_{\frac{\lambda\mathbf{M}}{\rho}}(\mathbf{Z}^{k+1} + \frac{\mathbf{T}_2^k}{\rho})$
5: Update $\mathbf{T}_1^{k+1}$and $\mathbf{T}_2^{k+1}$ as Eqs. (11)-(12)
6: $k = k + 1$
7: **if** ($\|\mathbf{Z}^k - \mathbf{S}^k\|_F < \varepsilon$ and $\|\mathbf{S}^k - \mathbf{S}^{k-1}\|_F < \varepsilon$or$k > \quad k_{\max}$)
8: Terminate=False
9: **end if**
10: **end while**
Output: Optimal sparse coefficient vector $\mathbf{S} = \mathbf{S}^k$

B. Semisupervised Sparse Manifold Hypergraphs

Based on the above-mentioned GNSR model, we can discover the intrinsic sparse and manifold properties of HSI data effectively. Considering pixels in HSI generally contain complex multivariate relationships, for instance, one point belongs to multiple manifold or sparse neighborhoods, while one neighborhood contains multiple points. Thus, a traditional simple graph just containing two vertices in one edge is unable to describe this higher order relationship.

HG is an effective tool to represent the multivariate relationship of HSI. In HG, a hyperedge is composed of multiple connected vertices, while a vertex belongs to multiple hyperedges, which exactly corresponds to the multivariate manifold or sparse structure in HSI. In the HIS dataset, the pixels belonging to the same class usually lie on the same nonlinear manifold subspace and have a stronger spare correlation. Therefore, we use the sparse coefficients and geodesic distances between unlabeled and labeled samples to construct an unsupervised sparse manifold HG $\mathbf{G}^u$. $\mathbf{G}^u$ represents the multivariate sparse relations between unlabeled samples lie on the same manifold and further preserves the HG structure to discover more potential discriminant information. Meanwhile, relying on the HG framework, a discriminant manifold HG $\mathbf{G}^l$ is designed for discovering the discriminant manifold structure of labeled samples.

For unlabeled samples, their probabilistic relationship with different classes is implied to discover the potential discriminant information. The coefficients obtained by the GNSR model reflect the sparse relationship and manifold similarities between unlabeled samples and labeled samples. Therefore, we adopt these sparse manifold coefficients to calculate the similarity relationship between unlabeled sample $\mathbf{x}_i^u$ and the tth class as

$$d_t\left(\mathbf{x}_i^u\right) = \frac{1}{n_t k}\sum_{\mathbf{x}_p^u \in N\left(\mathbf{x}_i^u\right)}\sum_{\mathbf{x}_q^l \in M(t)} s_{pq}, \quad t = 1, \ldots, c \quad (13)$$

in which $N(\mathbf{x}_i^u)$ is local neighborhood set of unlabeled sample $\mathbf{x}_i^u$ and $M(t)$ contains all labeled samples from tth class. k_g and n_t are the numbers of samples in $N(\mathbf{x}_i^u)$ and $M(t)$, respectively. s_{pq} is the calculated sparse manifold coefficient between samples $\mathbf{x}_p^u$ and $\mathbf{x}_q^l$. The similarity $d_t(\mathbf{x}_i^u)$ is the sum of sparse manifold coefficient between all neighbors of $\mathbf{x}_i^u$ and all labeled samples from tth class. A large $d_t(\mathbf{x}_i^u)$ denotes that not only $\mathbf{x}_i^u$ is similar with tth class but also its neighbors are similar with tth class. Thus, the prelabel of $\mathbf{x}_i^u$ can be set as

$$\hat{l}\left(\mathbf{x}_i^u\right) = \arg\max_{t=1,\ldots,c} d_t\left(\mathbf{x}_i^u\right). \quad (14)$$

Relying on the prelabels of unlabeled samples, the hyperedge weight $w(e_i^u)$ and incidence matrix $\mathbf{H}^u = [h(\mathbf{x}_i, e_j^u)]_{n,m}$of unsupervised HG $\mathbf{G}^u$ are defined as follows:

$$w\left(e_j^u\right) = \sum_{i=1}^{n} s_{ij}\exp\left(-\frac{n d_{\mathrm{geo}}\left(\mathbf{x}_i^u, \mathbf{x}_j^l\right)}{\sum_{p=1}^{n} d_{\mathrm{geo}}\left(\mathbf{x}_p^u, \mathbf{x}_j^l\right)}\right) \quad (15)$$

$$h\left(\mathbf{x}_i^u, e_j^u\right) = \begin{cases} 1, & \text{if } \hat{l}\left(\mathbf{x}_i^u\right) = l\left(\mathbf{x}_j^l\right) \text{ and } s_{ij} \neq 0 \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

Then, the degrees of hyperedge e_i^u and vertex $\mathbf{x}_i^u$ are obtained by

$$d\left(\mathbf{x}_i^u\right) = \sum_{j=1}^{n} h\left(\mathbf{x}_i^u, e_j^u\right) w\left(e_j^u\right), \quad d\left(e_j^u\right) = \sum_{i=1}^{n} h\left(\mathbf{x}_i^u, e_j^u\right). \quad (17)$$

According to (15) and (16), HG $\mathbf{G}^u$ sets total m hyperedges corresponding to m labeled sample, and each hyperedge e_i^u contains the unlabeled samples that not only have the same per-label but also belong to the sparse neighbors of $\mathbf{x}_i^l$, which ensures the unlabeled samples owing stronger sparse manifold similarities are connected in HG. To maintain the multivariate similarity relation between unlabeled samples, an objective function with unlabeled samples is constructed

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as (18), shown at the bottom of the page. Through some mathematical operations, the above-mentioned function is converted into (19), as shown at the bottom of the page, in which $\mathbf{L}_H^u$ is hyper-Laplacian matrices of HG $\mathbf{G}^u$, and $\mathbf{D}_x^u$, $\mathbf{D}_e^u$, and $\mathbf{W}^u$ are the diagonal matrices of $d(\mathbf{x}_i^u)$, $d(e_i^u)$, and $w(e_i^u)$, respectively.

Beside considering the unlabeled samples, the labeled samples are also utilized for training via a supervised HG $\mathbf{G}^l$. In $\mathbf{G}^l$, a hyperedge e_i^l of labeled sample $\mathbf{x}_i^l$ is constructed with intraclass samples, and the weight of e_i^l is defined as

$$w\left(e_j^l\right) = \sum_{i=1}^{m} \exp\left(-\frac{m d_{\text{geo}}\left(\mathbf{x}_i^l, \mathbf{x}_j^l\right)}{\sum_{p=1}^{m} d_{\text{geo}}\left(\mathbf{x}_j^l, \mathbf{x}_p^l\right)}\right). \tag{21}$$

The corresponding incidence matrix $\mathbf{H}^s = [h(\mathbf{x}_i, e_j^l)]_{i,j}$ is given as

$$h\left(\mathbf{x}_i^l, e_j^l\right) = \begin{cases} 1, & \text{if } l\left(\mathbf{x}_i^l\right) = l\left(\mathbf{x}_j^l\right) \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

With the above-mentioned hyperedge weight and incidence matrix, the degrees of hyperedge e_i^l and vertex $\mathbf{x}_i^l$ can be obtained as follows:

$$d\left(\mathbf{x}_i^l\right) = \sum_{j=1}^{m} h\left(\mathbf{x}_i^l, e_j^l\right) w\left(e_j^l\right), \quad d\left(e_j^l\right) = \sum_{i=1}^{m} h\left(\mathbf{x}_i^l, e_j^l\right). \tag{23}$$

To fully use the valuable discriminant information in training, the objective function (20), as shown at the bottom of the page, based on HG Gl is constructed by compacting intraclass samples as close as possible in low-dimensional space, in which $\mathbf{L}_H^l$ is hyper-Laplacian matrices of HG $\mathbf{G}^l$, while $\mathbf{D}_x^l$, $\mathbf{D}_e^l$, and $\mathbf{W}^l$ are the diagonal matrices of $d(\mathbf{x}_i^l)$, $d(e_i^l)$, and $w(e_i^l)$, respectively.

Based on the above-mentioned mathematical transformation, the objective function of GSMH can be further simplified as

$$\begin{cases} \min\limits_{\mathbf{V}} \ \text{tr}\left\{\mathbf{V}^{\mathrm{T}} \mathbf{X}_u \mathbf{L}_H^u \mathbf{X}_u^{\mathrm{T}} \mathbf{V}\right\} \\ \min\limits_{\mathbf{V}} \ \text{tr}\left\{\mathbf{V}^{\mathrm{T}} \mathbf{X}_l \mathbf{L}_H^l \mathbf{X}_l^{\mathrm{T}} \mathbf{V}\right\}. \end{cases} \tag{24}$$

To incorporate the labeled and unlabeled training information in embedding, an objective function of the semisupervised HG embedding framework is designed as the following:

$$\begin{cases} \min\limits_{\mathbf{V}} \ \text{tr}\left\{\mathbf{V}^{\mathrm{T}}\left[\alpha \mathbf{X}_l \mathbf{L}_H^l \mathbf{X}_l^{\mathrm{T}} + (1-\alpha)\mathbf{X}_u \mathbf{L}_H^u \mathbf{X}_u^{\mathrm{T}}\right]\mathbf{V}\right\} \\ \text{s.t.} \ \ \mathbf{V}^{\mathrm{T}} \mathbf{X}_u \mathbf{D}_x^u \mathbf{X}_u^{\mathrm{T}} \mathbf{V} = \mathbf{I}, \quad \mathbf{V}^{\mathrm{T}} \mathbf{X}_l \mathbf{D}_x^l \mathbf{X}_l^{\mathrm{T}} \mathbf{V} = \mathbf{I} \end{cases} \tag{25}$$

where the tradeoff parameter $\alpha \in [0, 1]$ balances the contribution of labeled and unlabeled samples in FE, and two constraint items are utilized to reduce sample scale influence in embedding.

With the Lagrangian multiplier method, the above-mentioned objective function is changed to solve the following generalized eigenvalue problem:

$$\begin{aligned} &\left[\alpha \mathbf{X}_l \mathbf{L}_H^l \mathbf{X}_l^{\mathrm{T}} + (1-\alpha)\mathbf{X}_u \mathbf{L}_H^u \mathbf{X}_u^{\mathrm{T}}\right]\mathbf{v}_i \\ &\qquad = \lambda_i \left(\mathbf{X}_u \mathbf{D}_x^u \mathbf{R} \mathbf{X}_u^{\mathrm{T}} + \mathbf{V} \mathbf{X}_l \mathbf{D}_x^l \mathbf{X}_l^{\mathrm{T}}\right)\mathbf{v}_i. \end{aligned} \tag{26}$$

The optimal projection matrix $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_d] \in \Re^{D \times d}$ consists of the smallest d eigenvalues corresponding eigenvectors, and the extracted features of HSI data can be denoted as $\mathbf{Y} = \mathbf{V}^{\mathrm{T}} \mathbf{X} \in \Re^{d \times (m+n)}$.

$$\min_{\mathbf{V}} \frac{1}{2} \sum_{e_k^u} \frac{w\left(e_k^u\right) h\left(\mathbf{x}_i^u, e_k^u\right) h\left(\mathbf{x}_j^u, e_k^u\right)}{d\left(e_k^u\right)} \sum_{\mathbf{x}_i^u, \mathbf{x}_j^u \in e_k^u} \left\| \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u}{\sqrt{d\left(\mathbf{x}_i^u\right)}} - \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_j^u}{\sqrt{d\left(\mathbf{x}_j^u\right)}} \right\|_2^2 \tag{18}$$

$$\begin{aligned} &\min_{\mathbf{V}} \frac{1}{2} \sum_{e_k^u} \frac{w\left(e_k^u\right) h\left(\mathbf{x}_i^u, e_k^u\right) h\left(\mathbf{x}_j^u, e_k^u\right)}{d\left(e_k^u\right)} \sum_{\mathbf{x}_i^u, \mathbf{x}_j^u \in e_k^u} \left\| \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u}{\sqrt{d\left(\mathbf{x}_i^u\right)}} - \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u}{\sqrt{d\left(\mathbf{x}_j^u\right)}} \right\|_2^2 \\ &= \min_{\mathbf{V}} \sum_{e_k^u} \sum_{\mathbf{x}_i^u, \mathbf{x}_j^u \in e_k^u} \frac{w\left(e_k^u\right) h\left(\mathbf{x}_i^u, e_k^u\right) h\left(\mathbf{x}_j^u, e_k^u\right)}{d\left(e_k^u\right)} \left(\frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u \left(\mathbf{x}_i^u\right)^{\mathrm{T}} \mathbf{V}}{d\left(\mathbf{x}_i^u\right)} - \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u \left(\mathbf{x}_j^u\right)^{\mathrm{T}} \mathbf{V}}{\sqrt{d\left(\mathbf{x}_i^u\right) d\left(\mathbf{x}_j^u\right)}} \right) \\ &= \min_{\mathbf{V}} \sum_{e_k^u} \sum_{\mathbf{x}_i^u \in e_k^u} \frac{w\left(e_k^u\right) h\left(\mathbf{x}_i^u, e_k^u\right) \mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u \left(\mathbf{x}_i^u\right)^{\mathrm{T}} \mathbf{V}}{d\left(e_k^u\right)} \sum_{\mathbf{x}_j^u \in e_k^u} \frac{h\left(\mathbf{x}_j^u, e_k^u\right)}{d\left(e_k^u\right)} - \sum_{e_k^u} \sum_{\mathbf{x}_i^u, \mathbf{x}_j^u \in e_k^u} \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u w\left(e_k^u\right) h\left(\mathbf{x}_i^u, e_k^u\right) h\left(\mathbf{x}_j^u, e_k^u\right) \left(\mathbf{x}_j^u\right)^{\mathrm{T}} \mathbf{V}}{d\left(e_k^u\right) \sqrt{d\left(\mathbf{x}_i^u\right) d\left(\mathbf{x}_j^u\right)}} \\ &= \min_{\mathbf{V}} \sum_{e_k^u} \sum_{\mathbf{x}_i^u \in e_k^u} \mathbf{V}^{\mathrm{T}} \mathbf{x}_i^u \left(\mathbf{x}_i^u\right)^{\mathrm{T}} \mathbf{V} - \text{tr}\left\{ \mathbf{V}^{\mathrm{T}} \mathbf{X}_u \left(\mathbf{D}_x^u\right)^{-\frac{1}{2}} \mathbf{H}^u \mathbf{W}^u \left(\mathbf{D}_e^u\right)^{-1} \left(\mathbf{H}^u\right)^{-T} \left(\mathbf{D}_x^u\right)^{-\frac{1}{2}} \mathbf{X}_u^{\mathrm{T}} \mathbf{V} \right\} \\ &= \text{tr}\left\{ \mathbf{V}^{\mathrm{T}} \mathbf{X}_u \left(\mathbf{I} - \left(\mathbf{D}_x^u\right)^{-\frac{1}{2}} \mathbf{H}^u \mathbf{W}^u \left(\mathbf{D}_e^u\right)^{-1} \left(\mathbf{H}^u\right)^{-T} \left(\mathbf{D}_x^u\right)^{-\frac{1}{2}} \right) \mathbf{X}_u^{\mathrm{T}} \mathbf{V} \right\} = \text{tr}\left\{ \mathbf{V}^{\mathrm{T}} \mathbf{X}_u \mathbf{L}_H^u \mathbf{X}_u^{\mathrm{T}} \mathbf{V} \right\} \end{aligned} \tag{19}$$

$$\begin{aligned} &\min_{\mathbf{V}} \frac{1}{2} \sum_{e_k^l} \frac{w\left(e_k^l\right) h\left(\mathbf{x}_i^l, e_k^l\right) h\left(\mathbf{x}_j^l, e_k^l\right)}{d\left(e_k^l\right)} \sum_{\mathbf{x}_i^l, \mathbf{x}_j^l \in e_k^l} \left\| \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_i^l}{\sqrt{d\left(\mathbf{x}_i^u\right)}} - \frac{\mathbf{V}^{\mathrm{T}} \mathbf{x}_j^l}{\sqrt{d\left(\mathbf{x}_j^u\right)}} \right\|_2^2 \\ &= \text{tr}\left\{ \mathbf{V}^{\mathrm{T}} \mathbf{X}_l \left(\mathbf{I} - \left(\mathbf{D}_x^u\right)^{-\frac{1}{2}} \mathbf{H}^u \mathbf{W}^u \left(\mathbf{D}_e^u\right)^{-1} \left(\mathbf{H}^u\right)^{-T} \left(\mathbf{D}_x^u\right)^{-\frac{1}{2}} \right) \mathbf{X}_l^{\mathrm{T}} \mathbf{V} \right\} = \text{tr}\left\{ \mathbf{V}^{\mathrm{T}} \mathbf{X}_l \mathbf{L}_H^l \mathbf{X}_l^{\mathrm{T}} \mathbf{V} \right\} \end{aligned} \tag{20}$$

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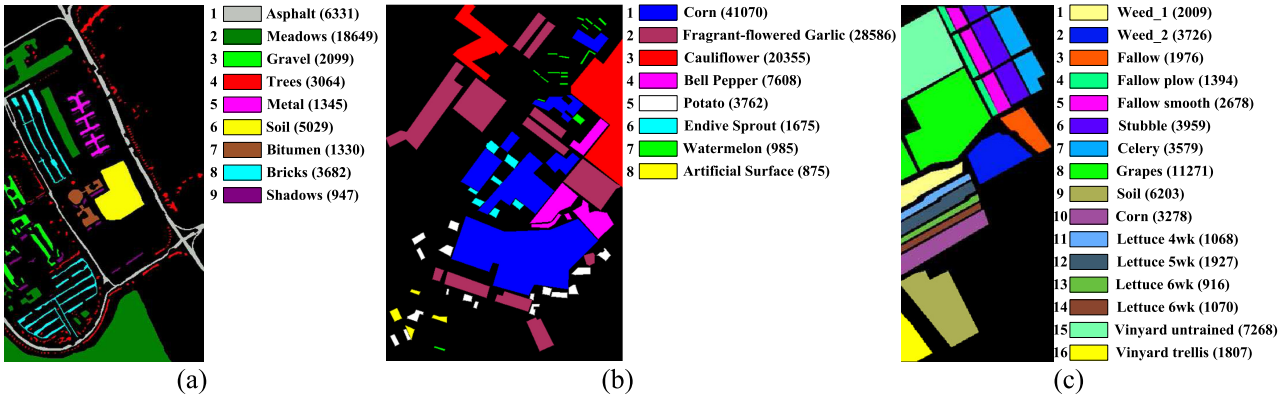


Fig. 3. Ground truth and class lists of three HSI datasets. (a) PaviaU. (b) Heihe. (c) Salinas.

III. Experimental Results and Analysis

To verify the performance of the GSMH method, a series of experiments are performed on three real hyperspectral datasets, and several state-of-art FE algorithms are compared with GSMH to demonstrate its effectiveness.

A. Datasets

PaviaU Dataset: It is a scene of the Pavia University, Pavia, Italy. It contains 640 × 340 pixels and 115 spectral bands over the range of 375–2500 nm. After removing absorption bands, 103 radiance channels remained for experiments. Nine classes of ground objects are marked in this sense. Its ground truth and classes list are exhibited in Fig. 3(a).

Heihe Dataset: It records the Zhangye basin located in the middle reaches of the Heihe watershed, Gansu Province, China. This dataset has 684 × 453 pixels, and its spatial resolution is up to 2.4 m. After the removal of 14 noise bands, 135 spectral bands are selected for the experiment. This dataset has a total of nine land cover types, and its ground truth with detailed class information is given in Fig. 3(b).

Salinas Dataset: This acquires the scene of Salinas Valley, Southern California, USA. The spatial size of this dataset is up to 512 × 217. Total 200 spectral channels without noise and water absorption are used in experiments. There are 16 land cover types in this dataset, and its ground truth with corresponding classes are displayed in Fig. 3(c).

B. Experimental Setup

To prove the superiority of the presented GSMH approach, we compared it and several FE algorithms, including three unsupervised methods PCA, LPP, and MSME, and six supervised methods LDA, LGSFA, SGDA, LSMDL, DHLP, and LCSMHL, and four semisupervised methods SDA, SELD, S^{3}MDA, and SHGE. The raw spectral feature (RSF) of HIS was exploited for comparison. The cross validation approach was adopted to optimize parameters in different algorithms. For LPP and LGSFA, the numbers of neighbors were set to 10. For the SR-based method MSME and SGDA, their sparse parameters were set to 5 and 1, respectively. The HG-based FE approaches MSME, DHLP, and SHGE all set the number of hyperedges equal to the training samples size. In semisupervised methods, SDA, SELD, and S^{3}MDA, the values of tradeoff parameters between the supervised and unsupervised items were set as 2, 1, and 5.

In experiments, an NN classifier, support vector machine (SVM), and random forest (RF) were used to classify the extracted features. The average classification accuracy of all classes (AA), overall classification accuracy (OA), and *kappa* coefficient (KC) were selected as evaluating indicators for experimental results.

C. Parameters and Dimension Selection

The proposed GSMH has two important parameters: the tradeoff parameter α balances the contribution of labeled and unlabeled samples during calculating projection matrix, while parameter λ controls the sparse degree in sparse optimization. In addition, feature dimension d is an important factor affecting the feature learning effectiveness of the FE method. To evaluate parameters and dimensions, 60 labeled samples per class and 1000 unlabeled samples were randomly selected for training, and the rest samples were tested by an NN classifier. Parameters λ and α were tuned with a set of $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}$ and a set of {0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1}, respectively. Feature dimension d was set with {5, 10, 15, 20, 25, 30, 35, 40, 45, 50}. The curved surface map of the OAs with changing α and λ are displayed in Fig. 4, while the OAs curves of FE methods with different feature dimensions were represented in Fig. 5.

According to Fig. 4, OAs are first significantly improved and then decreased with the growth of α. If α is too small, unlabeled samples will play a main role in the embedding process, while the prior information of labeled samples will be lost in embedding features, which directly reduces the discriminant power of extracted features. On the contrary, a too larger α will make GSMH pays more attention to labeled samples but ignores the sparse manifold similarities of unlabeled samples. The parameter λ shows a subtle change of OA in comparison with α. For increasing λ, OAs grow to a peak value since the GNSR model focuses on ℓ_1-norm spare item for selecting the suitable sparse neighbors for unlabeled samples. However, when λ keeps increasing, OAs will tend to

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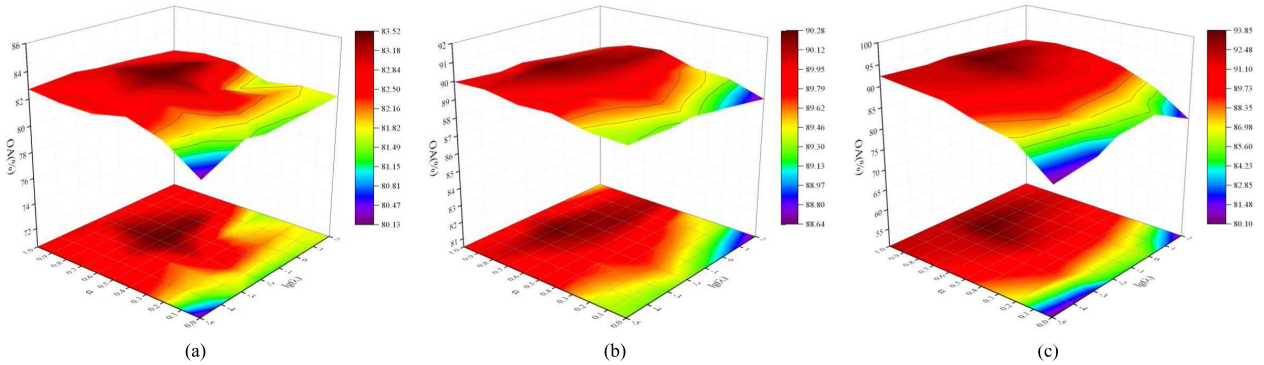


Fig. 4. Classification accuracies with different α and λ on three HSI datasets. (a) PaviaU. (b) Salinas. (c) Heihe.

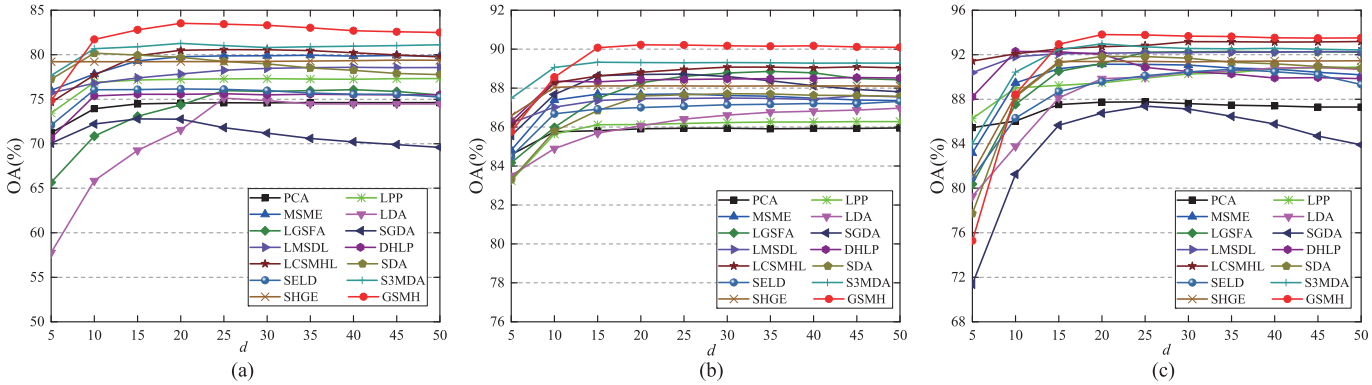


Fig. 5. Classification accuracies of FE methods with different feature dimensions d. (a) PaviaU. (b) Salinas. (c) Heihe.

decrease because a too large λ will result in selecting too few sparse neighbors to discover the real sparse property. In order to obtain optimal classification accuracy, α and λ are set to 0.6 and 10^{-2} for the PaviaU dataset, 0.8 and 10^{-2} for the Heihe and Salinas datasets, respectively.

As shown in Fig. 5, the HG-based approaches, MSME, DHLP, and SHGE, generally present better performance than other graph-based methods, such as LGSFA, SGDA, and LPP. These experimental results indicate that the HG revealing the multivariate relationships of samples are more effective than the traditional graph representing point-to-point relationships. S^{3}MDA, LCSMHL, and GSMH achieve the best performance among these FE methods due to considering the sparse and manifold properties simultaneously. The performance of LMSDL is slightly inferior to LCSMHL and GSMH for its graph framework is unable to represent the complex multivariate relationship of samples. Due to the proposed method is designed on preserving multivariate sparse and manifold structure of nonlinear data, too small dimensional feature space will limit the representation ability of these complex structures. Therefore, the proposed GMSH is slightly inferior to a few algorithms on the Heihe dataset when feature space is too small, while GSMH is still obviously superior to other supervised methods and semisupervised methods in most cases, which indicates the multivariate sparse manifold correlation preserved in embedding process is valuable for enhancing the representation power of embedding features. Based on the results in Fig. 5, we set the optimal feature dimension as 30 in the following experiments.

D. Investigate of SR Model and Hypergraph

For the proposed GNSR SR model and GSMH HG embedding algorithm, a series of ablation experiments is performed to verify the effectiveness of their innovation points. GNSR has two major improvements: 1) utilizing the geodesic distance weights to optimize sparse coefficients in sparse items. 2) In the SR error item, it considers the local nonlinear structure via the neighborhood reconstruction point. Therefore, we first remove the geodesic distance weights matrix $\mathbf{M}$ from GNSR as neighborhood SR (NSR) model and then further remove neighborhood reconstruction matrix $\mathbf{R}$ of NSR as traditional SR (TSR) model. Meanwhile, for exploring the effectiveness of geodesic distance, we replace the geodesic distance with the common Euclidean distance in the GNSR model to form the Euclidean distance-based neighborhood SR (ENSR) model.

The objective function of NSR is defined by

$$\begin{cases} \min\limits_{\mathbf{S}} \lambda\|\mathbf{S}\|_1 + \dfrac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\mathbf{R}\|_F^2 \\ \text{s.t.} \quad \mathbf{i}_m^{\mathrm{T}}\mathbf{Z} = \mathbf{i}_n, \ \mathbf{Z} = \mathbf{S}. \end{cases} \tag{27}$$

The objective function of the TSR model is given as follows:

$$\begin{cases} \min\limits_{\mathbf{S}} \lambda\|\mathbf{S}\|_1 + \dfrac{1}{2}\|\mathbf{X}_l\mathbf{Z} - \mathbf{X}_u\|_F^2 \\ \text{s.t.} \quad \mathbf{i}_m^{\mathrm{T}}\mathbf{Z} = \mathbf{i}_n, \ \mathbf{Z} = \mathbf{S}. \end{cases} \tag{28}$$

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TABLE I
CLASSIFICATION RESULTS OF GSMH WITH DIFFERENT HGS AND SR MODELS (OA ± STD)

Datasets		PaviaU		Heihe		Salinas	
SR	HG	NN	SVM	NN	SVM	NN	SVM
GNSR	GSMH	**83.52±0.98**	**86.87±1.49**	**93.81±0.90**	**95.32±1.13**	**90.27±1.18**	**90.43±1.14**
	GSMHws	82.69±1.28	85.29±1.09	93.69±0.90	94.86±0.54	89.69±1.11	90.23±0.93
	GSMHwp	80.85±1.02	84.08±1.54	92.75±0.81	93.75±0.35	87.12±0.73	89.22±0.90
ENSR	GSMH	82.37±1.25	85.16±1.22	92.93±0.96	94.33±1.07	88.41±1.29	89.89±1.39
	GSMHws	81.93±1.13	83.61±1.61	92.65±0.89	94.27±0.62	88.32±0.94	89.63±1.14
	GSMHwp	81.23±1.73	82.49±1.37	91.33±0.81	93.52±0.87	87.41±0.85	87.68±0.77
NSR	GSMH	81.83±1.01	85.25±1.73	92.62±0.83	94.88±1.11	88.67±1.17	90.12±1.12
	GSMHws	81.37±1.18	83.84±1.17	92.87±0.90	95.15±0.54	87.73±1.12	89.52±1.19
	GSMHwp	78.98±1.10	83.28±1.14	90.92±0.82	93.95±0.45	87.10±0.72	87.88±1.28
TSR	GSMH	80.61±1.53	83.78±1.32	92.51±0.74	94.16±1.25	87.24±1.21	89.73±1.56
	GSMHws	79.26±1.40	82.54±1.16	91.43±0.72	93.37±1.11	86.97±1.15	88.39±0.91
	GSMHwp	78.74±1.14	81.77±1.40	90.64±0.70	92.62±0.57	86.62±0.91	87.06±0.67

For the HG framework in GMSH, its improvements are mainly embodied in two aspects: 1) defining the hyperedges of unsupervised HG according to the sparse manifold similarities between unlabeled and labeled samples, rather than relationship only among unlabeled samples, which aims to fully incorporate labeled and unlabeled samples in embedding. 2) Computing the class probability of unlabeled sample and sets the incidence matrix of HG according to prelabel, in order to explore more potential discriminant information for training. Thus, we, respectively, remove the first and the second improvement to form new HGss "GSMH without sparse (GSMHws)" and "GSMH without prelabeled (GSMHwp)" HG embedding algorithms.

The hyperedge weight $w(e_i^u)$ and incidence matrix $\mathbf{H}^u = [h(\mathbf{x}_i, e_j^u)]_{n,m}$ of unsupervised HG in GSMHws are defined as

$$w\left(e_j^u\right) = \sum_{i=1}^{n} \exp\left(-\frac{n d_{\text{geo}}\left(\mathbf{x}_i^u, \mathbf{x}_j^u\right)}{\sum_{p=1}^{n} d_{\text{geo}}\left(\mathbf{x}_p^u, \mathbf{x}_j^u\right)}\right) \tag{29}$$

$$h\left(\mathbf{x}_i^u, e_j^u\right) = \begin{cases} 1, & \text{if } \hat{l}\left(\mathbf{x}_i^u\right) = \hat{l}\left(\mathbf{x}_j^u\right) \\ 0, & \text{otherwise.} \end{cases} \tag{30}$$

For unsupervised HG in GSMHwp, its hyperedge weight $w(e_i^u)$ is the same with GSMH. While the incidence matrix $\mathbf{H}^u = [h(\mathbf{x}_i, e_j^u)]_{n,m}$ of GSMHwp is set as

$$h\left(\mathbf{x}_i^u, e_j^u\right) = \begin{cases} 1, & \text{if } s_{ij} \neq 0 \\ 0, & \text{otherwise.} \end{cases} \tag{31}$$

Based on the above-mentioned GNSR, ENSR, NSR, and TSR models and GSMH, GSMHws, and GSMHwp HGs, we combined each SR model with each HG to obtain a total of nine different FE methods. For analyzing the effectiveness of each innovation point, we perform nine FE methods on a training set, including 1000 unlabeled samples and 60 labeled samples from each class, and NN and SVM are applied for classification. These experiments are repeated ten times in each condition. The OAs with standard deviation (STD) of different algorithms are displayed in Table I.

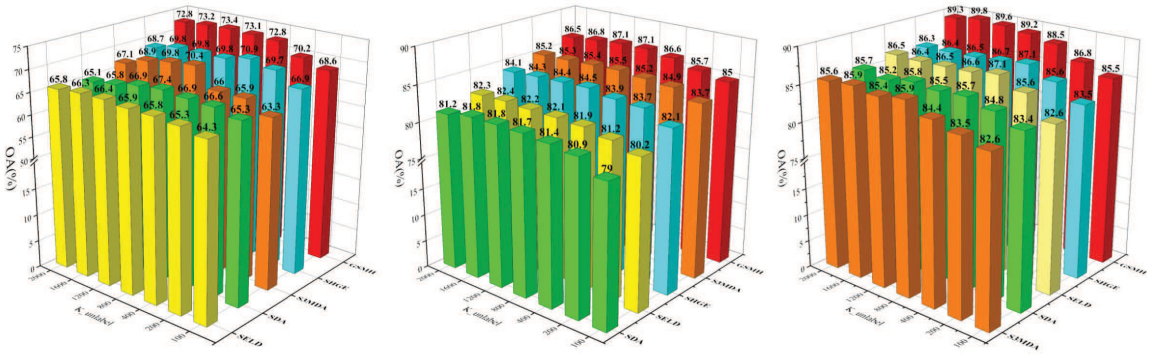
As shown in Table I, combining with the same HG and classifier, NSR generally presents better performance than TSR for considering the local manifold neighborhood information. The original GNSR model is significantly superior to the NSR and TSR models because of adjusting sparse coefficients via geodesic distance weights, which makes the obtained sparse coefficients reflect a more correct similarity relationship between labeled and unlabeled samples. The GNSR model using geodesic distance generally achieves better classification results than the ENSR model using the Euclidean distance no matter combined with which HGs. These experimental results obtained on three HSI datasets provide strong evidence that geodesic distance can effectively reveal the sparse correlations and manifold structure in nonlinear feature space. For HGs, GSMHws achieves better results in comparison with GSMHwp due to using prelabels of unlabeled samples to explore more potential prior discriminant information for training. Among three HGs, the original GSMH exhibits the best performance on all datasets. These experimental results indicate that the semisupervised incorporation information and class probabilistic relationship used in HG of GSMH are beneficial to reveal the discriminant property of HSI data.

E. Analysis of Training Conditions

To investigate the influence of unlabeled samples number on classification accuracy, K_u unlabeled samples and ten labeled samples from each class were selected for training. The value of K_u was, respectively, set with 100, 200, 400, 800, 1200, 1600, and 2000, and NN classifier was used to classify the remaining samples. In experiments, GSMH was compared with four semisupervised methods SDA, SELD, SHGE, and S^{3}MDA, and their classification accuracies on three datasets are shown in Fig. 6.

According to Fig. 6, as the increase of unlabeled sample size, the classification accuracies of all semisupervised methods first grow greatly and then reach stable values for unlabeled information tends to saturation. S^{3}MDA considers both sparse and manifold structures in graphs, while SHGE reveals the multivariate similarity relationship via HG. Therefore, they generally present better performance than SDA and SELD in most cases. GSMH is obviously superior to S^{3}MDA and SHGESHGE because it not only considers the multivariate sparse manifold correlation in HGs but also discovers the discriminant relation between unlabeled and labeled training sets.

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(a) (b) (c)

Fig. 6. Classification accuracies of semisupervised FE methods with different unlabeled training sets. (a) PaviaU. (b) Salinas. (c) Heihe.

TABLE II
CLASSIFICATION RESULTS OF DIFFERENT FE METHOD WITH NN CLASSIFIER ON PAVIAU DATASET [OA ± STD (KAPPA)]

K_l	5	10	20	40	60
RSF	62.04±3.30(0.53)	62.36±2.07(0.54)	67.88±1.63(0.60)	69.09±0.92(0.62)	73.65±0.43(0.67)
PCA	63.04±3.31(0.54)	63.35±2.07(0.55)	69.88±1.62(0.62)	70.05±0.93(0.63)	74.62±0.42(0.68)
LPP	64.36±3.01(0.55)	66.13±1.81(0.58)	72.58±1.55(0.65)	73.75±1.22(0.67)	77.32±0.74(0.71)
MSME	65.02±2.30(0.56)	68.76±1.39(0.61)	75.55±0.92(0.69)	77.11±0.75(0.71)	79.94±0.45(0.74)
LDA	48.66±3.12(0.37)	54.14±2.43(0.44)	60.03±1.45(0.50)	70.82±0.76(0.63)	75.13±0.75(0.68)
LGSFA	55.66±3.72(0.45)	56.33±1.64(0.46)	64.94±1.62(0.56)	72.76±0.90(0.65)	76.07±0.72(0.69)
SGDA	65.55±2.38(0.57)	69.71±1.81(0.62)	72.19±1.81(0.65)	73.11±0.92(0.66)	73.91±0.81(0.66)
LMSDL	62.05±2.78(0.53)	68.61±7.82(0.61)	75.48±1.94(0.69)	78.42±2.97(0.72)	78.55±1.37(0.72)
DHLP	62.45±2.06(0.53)	67.58±1.90(0.59)	70.35±1.33(0.63)	72.13±0.91(0.65)	75.61±0.85(0.69)
LCSMHL	58.57±3.65(0.48)	65.53±3.07(0.56)	74.38±2.44(0.68)	78.23±1.59(0.72)	79.84±0.88(0.73)
SDA	65.31±3.31(0.57)	65.86±2.08(0.58)	73.37±1.82(0.66)	75.43±0.65(0.69)	80.18±0.23(0.75)
SELD	65.03±3.29(0.56)	65.75±1.98(0.57)	70.89±1.59(0.63)	73.73±0.59(0.67)	76.16±0.43(0.70)
S^3MDA	67.85±2.77(0.59)	70.11±1.58(0.62)	73.05±1.51(0.66)	76.52±1.43(0.70)	80.62±0.81(0.75)
SHGE	67.89±2.09(0.59)	72.79±1.12(0.65)	75.15±1.05(0.68)	76.56±0.53(0.70)	79.39±0.33(0.73)
GSMH	**70.14±2.83(0.62)**	**73.31±1.45(0.66)**	**77.69±0.85(0.71)**	**80.76±0.63(0.75)**	**83.52±0.51(0.78)**

Note: The best results of a column are marked in bold.

TABLE III
CLASSIFICATION RESULTS OF DIFFERENT FE METHOD WITH NN CLASSIFIER ON SALINAS DATASET [OA ± STD (KAPPA)]

K_l	5	10	20	40	60
RSF	77.92±1.56(0.76)	80.66±1.02(0.79)	82.42±0.94(0.80)	83.81±0.62(0.82)	84.97±0.32(0.83)
PCA	78.92±1.55(0.77)	81.67±1.03(0.80)	83.38±0.92(0.81)	84.81±0.61(0.83)	85.95±0.31(0.84)
LPP	80.14±1.60(0.78)	83.35±0.84(0.82)	84.67±0.81(0.83)	85.26±0.66(0.84)	86.28±0.32(0.85)
MSME	81.01±2.68(0.79)	84.73±1.15(0.83)	86.33±0.62(0.84)	86.85±0.59(0.85)	87.36±0.48(0.86)
LDA	74.15±3.23(0.71)	65.67±2.52(0.62)	83.49±0.74(0.82)	86.29±0.68(0.85)	86.97±0.26(0.85)
LGSFA	80.12±3.31(0.78)	83.56±0.94(0.82)	85.27±0.64(0.84)	87.17±0.58(0.86)	88.43±0.50(0.87)
SGDA	79.79±2.26(0.78)	83.43±0.95(0.82)	86.26±0.62(0.85)	88.09±0.63(0.87)	87.81±0.56(0.86)
LMSDL	80.36±1.49(0.78)	82.16±1.25(0.80)	84.85±1.07(0.83)	86.92±0.89(0.85)	87.31±0.87(0.86)
DHLP	81.16±2.68(0.08)	85.87±0.89(0.84)	87.39±0.85(0.86)	88.23±0.79(0.88)	88.52±0.58(0.88)
LCSMHL	81.32±1.19(0.79)	83.96±0.74(0.82)	86.56±2.85(0.85)	88.58±0.47(0.88)	88.94±0.69(0.88)
SDA	78.12±1.35(0.76)	81.32±1.07(0.79)	84.46±0.85(0.83)	86.73±0.48(0.85)	87.57±0.46(0.86)
SELD	78.29±1.79(0.76)	81.51±1.11(0.79)	84.79±0.72(0.83)	86.67±0.55(0.85)	87.32±0.54(0.86)
S^3MDA	82.21±3.65(0.81)	85.63±1.44(0.84)	87.10±1.16(0.86)	89.02±1.05(0.88)	88.98±0.57(0.88)
SHGE	81.12±2.26(0.79)	84.68±1.14(0.83)	86.51±1.01(0.85)	87.66±0.80(0.86)	88.12±0.48(0.87)
GSMH	**82.63±2.26(0.81)**	**86.64±1.13(0.85)**	**87.83±0.97(0.86)**	**89.63±0.58(0.88)**	**90.29±0.57(0.89)**

For exploring the performance with different labeled training samples sizes, we also chose 1000 unlabeled samples and K_l labeled samples per class as the training set. K_l was set as 5, 10, 20, 40, and 60, respectively. In this experiment, we compare GSMH with all the other FE methods, including three unsupervised methods, four supervised methods, and four semisupervised methods. Each experiment was repeated ten times, and NN was used for classifying. Tables II–IV give the average OAs with STD and KCs of FE methods on three HSI datasets, respectively.

As shown in Tables II–IV, PCA and LPP as unsupervised methods present unsatisfactory performance without using the *a priori* labeled information in HSI. LSMDL and LCSMHL are better than other supervised methods because they simultaneously reveal the nonlinear manifold structure and sparse relationship of data. Semisupervised methods S^3MDA, SHGE,

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT FE METHOD WITH NN CLASSIFIER ON HEIHE DATASET [OA ± STD (KAPPA)]

K_l	5	10	20	40	60
RSF	79.34±3.93(0.73)	80.46±3.83(0.74)	85.64±0.82(0.81)	87.75±0.82(0.84)	87.84±0.77(0.84)
PCA	79.33±3.92(0.73)	80.45±3.81(0.74)	85.63±0.82(0.81)	87.74±0.82(0.84)	87.77±0.76(0.84)
LPP	82.93±4.22(0.77)	84.38±2.13(0.79)	88.32±1.47(0.84)	90.14±0.85(0.87)	90.88±0.79(0.88)
MSME	81.91±4.26(0.76)	85.92±2.64(0.81)	88.68±1.07(0.85)	90.31±0.76(0.87)	91.14±0.73(0.88)
LDA	69.19±4.78(0.61)	69.55±3.04(0.61)	72.52±2.11(0.64)	91.18±1.12(0.88)	91.68±0.52(0.89)
LGSFA	77.31±3.66(0.71)	82.43±1.98(0.77)	90.39±1.53(0.87)	92.16±0.75(0.90)	92.56±0.64(0.91)
SGDA	81.72±2.79(0.76)	85.27±1.81(0.80)	87.07±1.28(0.83)	87.51±1.21(0.83)	87.38±0.84(0.84)
LMSDL	80.41±4.05(0.75)	85.93±3.02(0.81)	88.17±1.83(0.84)	91.71±0.88(0.89)	92.61±0.62(0.91)
DHLP	82.04±1.72(0.76)	86.80±1.72(0.82)	90.92±0.88(0.88)	91.52±0.81(0.89)	92.30±0.74(0.90)
LCSMHL	82.65±3.22(0.77)	86.15±1.97(0.81)	89.21±1.18(0.87)	92.85±0.79(0.91)	93.16±0.66(0.92)
SDA	81.54±3.71(0.76)	84.81±3.03(0.80)	88.94±1.58(0.85)	91.17±0.75(0.88)	91.85±0.74(0.89)
SELD	82.57±3.68(0.77)	86.52±2.31(0.82)	90.11±1.19(0.87)	90.51±1.03(0.87)	90.69±0.79(0.88)
S^3MDA	80.64±4.93(0.75)	85.33±2.75(0.80)	88.94±1.45(0.85)	91.51±0.79(0.89)	92.98±0.71(0.91)
SHGE	82.17±4.22(0.76)	86.34±2.32(0.82)	88.07±2.17(0.84)	90.31±0.88(0.87)	91.45±0.87(0.88)
GSMH	**86.15±2.75(0.82)**	**89.19±2.04(0.86)**	**91.67±0.92(0.89)**	**92.74±0.71(0.90)**	**93.81±0.41(0.92)**

TABLE V
CLASSIFICATION RESULTS OF FE METHODS WITH DIFFERENT CLASSIFIERS ON THREE HSI DATASET (OA ± STD)

Datasets	PaviaU		Salias		Heihe	
Methods	SVM	RF	SVM	RF	SVM	RF
RSF	74.37±1.81	75.02±2.48	88.17±1.12	86.27±0.82	91.95±0.44	91.12±0.16
PCA	77.46±1.65	77.52±1.14	89.66±1.31	87.13±0.57	91.94±0.44	92.81±0.80
LPP	77.85±1.82	77.63±1.61	89.69±0.61	88.91±0.24	92.73±0.41	92.82±1.14
MSME	75.52±1.63	80.58±1.86	90.23±0.48	89.36±0.45	93.89±1.18	93.69±0.66
LDA	75.71±1.02	75.22±0.73	89.30±0.25	87.84±1.17	92.42±0.41	92.67±0.74
LGSFA	78.04±2.10	80.19±1.72	90.21±0.98	89.79±0.31	93.79±0.57	93.13±0.72
SGDA	77.41±2.69	72.27±1.16	88.55±0.81	88.55±0.68	91.46±0.61	87.38±1.74
LMSDL	80.34±2.41	76.42±1.71	90.31±1.22	88.61±1.04	92.62±0.78	92.14±1.27
DHLP	79.81±1.74	78.65±1.58	90.24±1.14	89.27±0.47	94.35±0.48	93.28±0.82
LCSMHL	82.07±1.46	79.02±1.34	89.72±0.81	90.41±0.95	93.71±0.91	92.53±1.01
SDA	82.35±1.62	80.72±1.63	89.92±0.50	89.71±0.62	93.65±1.14	93.54±0.33
SELD	80.24±1.72	77.26±0.90	90.83±0.42	88.58±1.25	93.09±0.84	93.64±0.66
S^3MDA	83.63±2.21	81.82±1.02	90.21±0.33	90.01±0.65	94.54±0.78	93.65±0.51
SHGE	81.22±2.36	77.51±1.42	**90.82±0.37**	90.44±0.59	93.34±1.64	92.97±0.45
GSMH	**86.87±1.90**	**84.29±0.87**	90.43±0.75	**91.08±0.91**	**95.32±1.02**	**93.87±0.72**

TABLE VI
CLASSIFICATION RESULTS OF EACH CLASS WITH NN CLASSIFIER ON PAVIAU DATASET

Class	1	2	3	4	5	6	7	8	9	OA	AA	Time(s)
RSF	70.1±1.5	71.5±1.7	69.8±1.5	89.3±0.7	98.9±0.1	67.1±1.7	89.8±0.9	73.5±1.2	99.9±0.0	74.1±0.4	81.1±1.3	0.0±0.0
PCA	72.1±2.2	72.2±1.2	69.8±2.5	92.8±0.4	99.4±0.0	66.4±1.8	91.3±0.5	68.1±1.7	99.9±0.0	74.5±0.4	81.3±1.3	0.1±0.0
LPP	75.2±2.9	74.8±3.5	74.5±3.7	87.2±1.7	99.5±0.1	75.5±2.7	89.7±0.8	72.6±3.7	99.9±0.0	77.3±0.7	83.2±2.3	0.2±0.0
MSME	71.0±2.0	78.9±1.8	77.6±1.6	89.3±0.6	99.8±0.0	88.5±0.6	92.5±0.4	72.3±1.3	99.4±0.0	80.4±0.5	85.6±1.2	13.7±0.6
LDA	61.8±3.3	78.6±2.4	70.5±2.3	93.4±0.8	**99.9±0.0**	75.8±3.3	69.9±3.2	57.2±6.2	99.5±0.1	75.3±0.8	78.5±2.5	0.1±0.0
LGSFA	60.2±5.3	78.4±2.3	71.4±2.5	93.4±0.6	99.7±0.0	80.3±1.7	83.1±1.9	60.7±4.9	99.9±0.0	76.2±0.7	80.7±1.6	3.9±0.3
SGDA	60.2±5.2	76.5±2.5	75.2±2.7	70.2±3.3	80.1±2.2	76.5±2.3	87.2±1.3	62.6±4.5	95.4±0.7	73.1±0.8	76.2±2.4	10.9±0.9
LMSDL	64.8±6.5	81.2±4.9	67.8±8.0	92.9±1.6	99.6±0.1	76.2±3.8	80.9±4.2	74.3±5.6	99.5±0.1	78.5±1.4	81.9±4.7	4.1±0.6
DHLP	53.3±6.0	80.7±2.2	72.6±2.4	94.5±0.8	99.8±0.0	79.5±2.1	86.7±1.9	56.3±4.6	95.3±0.7	75.8±0.9	79.8±1.8	9.4±0.8
LCSMHL	63.6±4.5	86.3±1.3	**80.6±1.8**	94.9±0.5	98.0±0.2	84.6±1.9	91.3±0.9	52.5±6.1	99.9±0.0	80.8±0.9	80.8±2.5	84.6±7.4
SDA	75.4±0.6	77.5±0.9	73.8±0.7	92.4±0.3	99.8±0.0	82.9±0.5	92.8±0.2	76.8±1.0	99.9±0.0	80.2±0.2	85.7±0.4	6.1±0.1
SELD	75.6±1.3	74.6±1.5	69.2±2.2	94.6±0.4	99.3±0.1	76.3±2.0	85.1±0.8	62.5±2.5	99.8±0.1	76.7±0.4	81.9±1.5	8.3±0.4
S^3MDA	71.5±2.6	78.5±2.2	72.4±3.9	94.5±0.7	99.3±0.1	88.2±1.4	92.2±0.8	73.4±3.3	99.9±0.0	80.9±0.8	86.1±1.9	20.8±1.7
SHGE	73.4±1.0	79.6±0.9	78.7±1.2	84.9±0.7	99.2±0.0	74.2±1.0	87.2±0.6	69.1±1.3	99.9±0.0	78.7±0.3	82.9±0.8	16.4±0.5
GSMH	**75.9±1.5**	**88.3±0.7**	77.6±1.7	**95.0±0.4**	99.5±0.0	**88.7±1.1**	**93.3±0.5**	**73.9±2.0**	**100±0.0**	**83.7±0.5**	**86.5±1.0**	168.3±8.6

Note: Experimental results reserve a decimal fraction, and the best results of a column are marked in bold.

and GSMH are obviously superior to other supervised and unsupervised methods, especially under small labeled training conditions. GSMH achieves the best classification results among all FE methods in most conditions, which benefits from using the probabilistic relationship between unlabeled samples and classes to further enhance the discriminant of extracted features.

F. Classification Performance

To analyze the adaptability of the proposed method with different classifiers, we combined FE methods with NN, SVM, and RF for classification. We select 60 labeled samples per class and a total of 1000 unlabeled samples for training and repeat each experiment ten times. Due to the classification results with NN classifier have been shown in the

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TABLE VII
CLASSIFICATION RESULTS OF EACH CLASS WITH NN CLASSIFIER ON HEIHE DATASET

Class	1	2	3	4	5	6	7	8	OA	AA	Time(s)
RSF	87.1±1.6	88.2±2.0	87.1±1.9	81.5±3.0	90.5±1.1	85.8±1.8	90.2±2.2	97.1±0.6	87.2±0.8	88.4±2.2	0.0±0.0
PCA	88.4±2.2	90.2±1.6	86.4±2.2	78.8±3.7	88.9±2.0	90.3±1.8	85.8±3.0	97.1±0.5	87.9±0.8	88.2±1.7	0.1±0.0
LPP	90.5±1.2	91.9±1.8	89.8±1.2	91.7±1.9	96.7±0.7	85.3±2.1	92.5±1.6	96.9±0.4	91.0±0.8	91.9±1.6	0.9±0.1
MSME	90.2±1.1	92.5±0.9	91.7±1.8	88.3±1.8	97.1±0.5	95.6±0.7	91.9±1.1	98.9±0.1	91.4±0.7	93.3±1.2	21.4±1.6
LDA	88.5±1.4	95.4±0.4	93.2±0.6	91.7±1.3	96.9±0.3	88.3±1.4	92.8±0.9	97.9±0.2	91.9±0.5	93.1±0.6	1.3±0.1
LGSFA	90.9±1.0	93.3±0.8	93.5±0.7	91.4±1.1	97.7±0.3	96.3±0.4	93.6±0.6	98.0±0.3	92.5±0.6	94.3±0.7	6.9±0.4
SGDA	90.1±2.3	82.2±3.1	87.4±2.2	93.1±1.5	83.6±3.0	89.1±1.8	88.1±2.1	94.2±1.2	87.4±0.8	88.5±1.8	18.3±1.5
LMSDL	90.2±1.7	96.6±0.6	92.3±0.9	90.3±1.5	97.4±0.5	92.6±1.1	95.6±0.5	95.7±0.6	92.7±0.6	93.8±1.0	10.4±0.6
DHLP	91.0±1.7	95.3±0.9	89.1±1.8	93.1±1.2	97.2±0.4	87.9±1.7	94.6±0.7	98.7±0.3	92.3±0.7	93.5±1.3	16.4±1.2
LCSMHL	91.1±1.0	94.8±0.5	92.6±1.0	92.8±1.2	96.7±0.6	94.6±1.0	95.9±0.5	99.1±0.2	93.3±0.7	94.9±0.7	156.4±10.3
SDA	90.9±1.7	93.7±1.0	92.2±1.7	90.5±1.8	97.0±0.5	**96.7±0.7**	93.5±1.0	98.8±0.3	92.3±0.7	94.1±1.2	13.2±1.0
SELD	86.6±2.4	**96.9±0.6**	90.2±2.2	86.8±1.8	98.4±0.3	89.8±2.1	91.9±1.1	98.9±0.2	91.0±0.8	92.5±1.5	23.7±1.9
S^3MDA	92.9±1.5	93.6±1.0	92.2±0.9	89.4±1.3	98.2±0.2	91.2±1.3	92.3±1.3	97.5±0.4	92.6±0.7	93.2±0.7	37.2±2.6
SHGE	90.2±2.1	94.5±1.2	90.4±2.2	91.2±2.3	95.9±0.8	88.9±2.3	92.8±1.4	98.8±0.2	91.7±0.9	92.8±1.5	28.5±2.5
GSMH	**92.9±0.8**	95.2±0.5	**94.3±0.4**	**93.3±0.5**	**98.2±0.2**	94.8±0.4	**97.5±0.3**	**99.8±0.0**	**94.1±0.4**	**95.7±0.4**	253.3±10.4

Note: Experimental results reserve a decimal fraction, and the best results of a column are marked in bold.

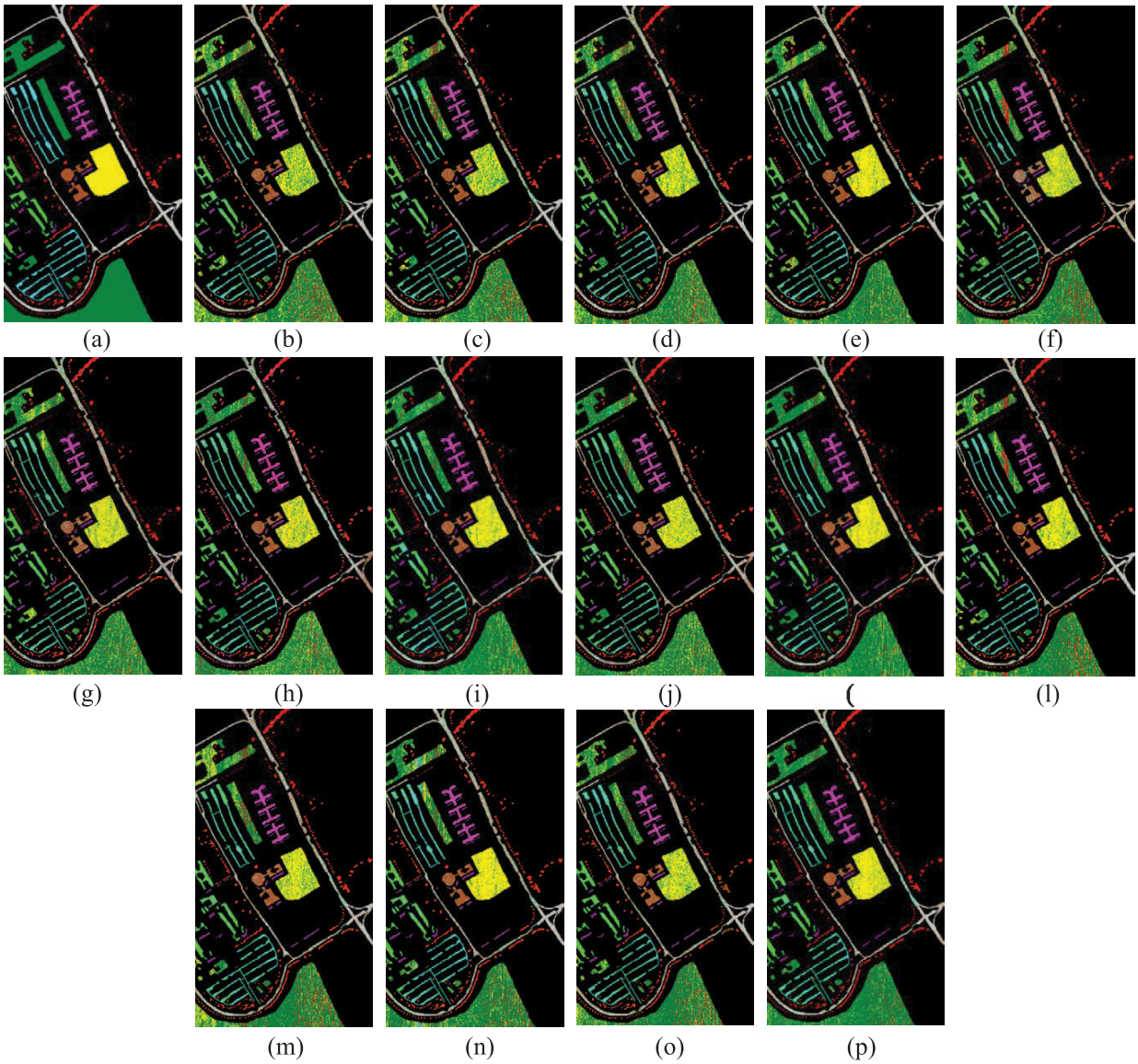


Fig. 7. Classification maps of different methods on PaviaU dataset. (a) GT. (b) RSF. (c) PCA). (d) LPP. (e) MSME. (f) LDA. (g) LGSFA. (h) SGDA. (i) LMSDL. (j) DHLP. (k) LCSMHL. (l) SDA. (m) SELD. (n) S^3MDA. (o) SHGE. (p) GSMH.

above-mentioned Tables II–IV, we only give classification results of SVM and RF on three HSI datasets in Table V. From Table V, we can clearly see that the proposed method achieves the best classification results with any classifiers in most conditions. The reason is that GSMH reveals the manifold-based sparse structure via the GNSR model and

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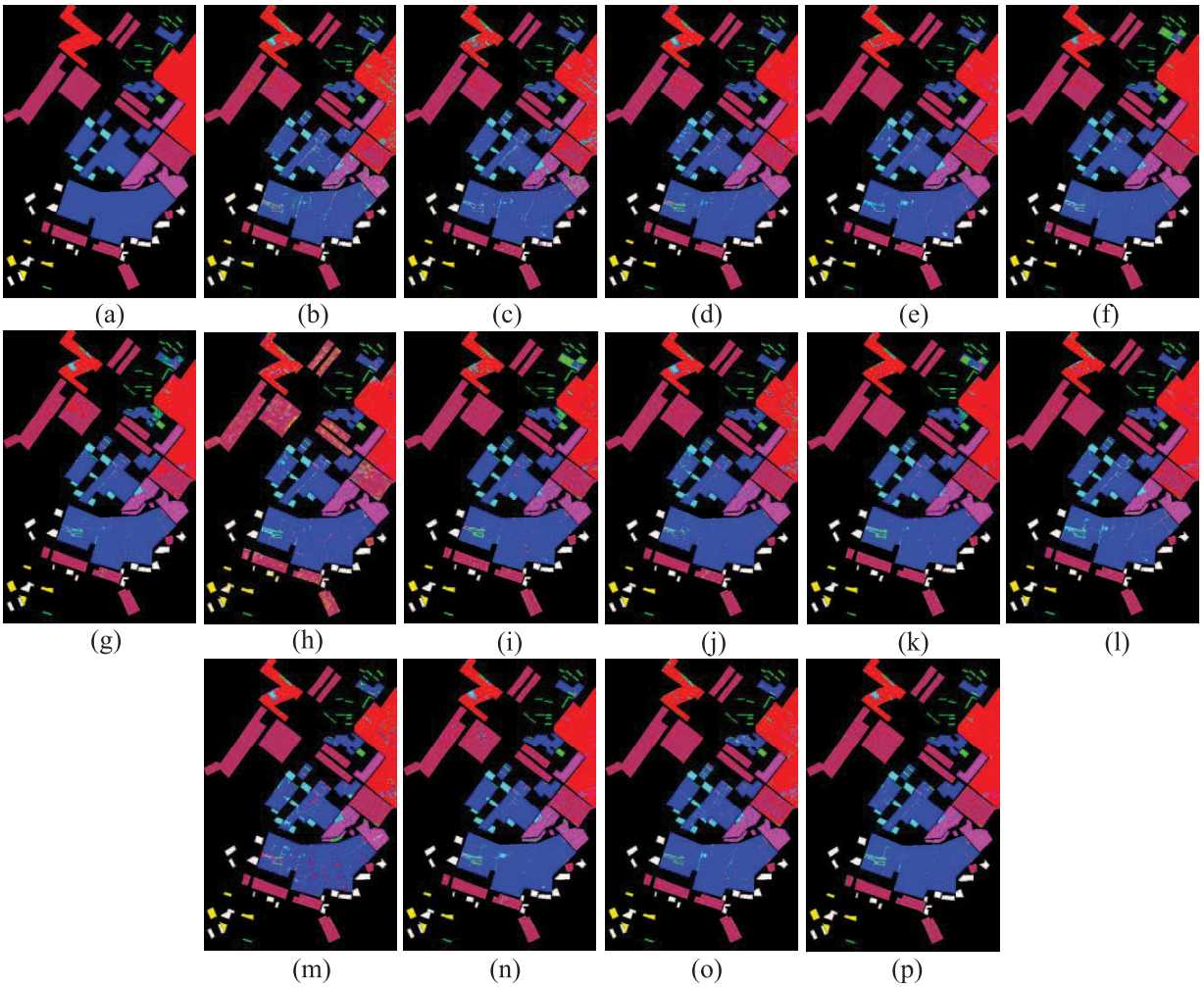


Fig. 8. Classification maps of different methods on Heihe dataset. (a) GT. (b) RAW. (c) PCA. (d) LPP. (e) MSME. (f) LDA. (g) LGSFA. (h) SGDA. (i) LMSDL. (j) DHLP. (k) LCSMHL. (l) SDA. (m) SELD. (n) S^3MDA. (o) SHGE. (p) GSMH.

enhances the intraclass compactness using sparse manifold HG. Therefore, the features obtained with GSMH contained essential structure properties and extensive discriminant information of original HSI data, which is beneficial for classifiers to distinguish unknown samples.

For verifying the classification performance of GSMH on different classes, all FE methods with NN classifier are performed under the same conditions as the above-mentioned experiment. The OA, AA, kappa, and classification accuracy of each class on the PaviaU and Heihe datasets are exhibited in Tables VI and VII, and the running times of FE processing are also shown in Tables VI and VII. The corresponding classification maps of FE methods are displayed in Figs. 7 and 8.

As shown in Tables VI and VII, the proposed GSMH achieves the best OA, AA, and KC among all FE approaches. Meanwhile, it obtains the highest classification accuracies in seven classes on the PaviaU dataset and six classes on the Heihe dataset. Due to computing geodesic distances between training samples, GSMH costs more computing times compared with other FE algorithms. However, relative to the negative influence of computing time growth, the significant improvement in classification accuracy of GSMH is more meaningful for distinguishing the land cover types.

In Figs. 7 and 8, the corresponding classification maps show that GSMH produces more homogenous areas and less misclassification points in comparison with other FE methods, especially in the regions of *Meadows*, *Soil*, and *Bitumen* on the PaviaU dataset and *Corn*, *Cauliflower*, and *Watermelon* on the Heihe dataset. It is clearly true that the proposed GSMH is effective to improve the discriminant representation power of embedding features and enhance the HSI classification performance under a small labeled training set.

IV. Conclusion

In this article, we presented a semisupervised FE method termed GSMH algorithm for HSI. GSMH first designs a GNSR model to reveal the sparse relation between different manifold neighborhoods and calculate the sparse coefficients reflecting nonlinear similarities between unlabeled and labeled samples. With the sparse coefficients and geodesic distance, GSMH constructs a pair of semisupervised HGs to incorporate unlabeled and labeled samples information in the training process and maintain the multivariate sparse manifold relation

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in embedding space. Meanwhile, GSMH considers the probabilistic relationship between unlabeled samples and each class in HG construction to discover more useful discriminant information, which further improves the feature discriminant power with limited labeled training samples. As a result, an optimal projection matrix is learned to extract useful discriminating features for improving HSI classification performance. The experimental results on three real-hyperspectral datasets demonstrate that the extracted feature possesses stronger feature representation power and better discriminant separability. Our future work will focus on using the spatial property of HIS to modify the proposed FE method.

REFERENCES

[1] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[2] R. Qureshi, M. Uzair, K. Khurshid, and H. Yan, "Hyperspectral document image processing: Applications, challenges and future prospects," *Pattern Recognit.*, vol. 90, pp. 12–22, Jun. 2019.

[3] L. Zhang, L. Zhang, B. Du, J. You, and D. Tao, "Hyperspectral image unsupervised classification by robust manifold matrix factorization," *Inf. Sci.*, vol. 485, pp. 154–169, Jun. 2019.

[4] H. Su, Y. Yu, Q. Du, and P. Du, "Ensemble learning for hyperspectral image classification using tangent collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3778–3790, Jun. 2020.

[5] O. Okwuashi and C. E. Ndehedehe, "Deep support vector machine for hyperspectral image classification," *Pattern Recognit.*, vol. 103, Jul. 2020, Art. no. 107298.

[6] W. He, H. Zhang, L. Zhang, W. Philips, and W. Liao, "Weighted sparse graph based dimensionality reduction for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 686–690, May 2016.

[7] A. Appice, P. Guccione, and D. Malerba, "A novel spectral-spatial co-training algorithm for the transductive classification of hyperspectral imagery data," *Pattern Recognit.*, vol. 63, pp. 229–245, Mar. 2017.

[8] Y. Dong, B. Du, L. Zhang, and L. Zhang, "Dimensionality reduction and classification of hyperspectral images using ensemble discriminative local metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2509–2524, May 2017.

[9] L. Zhang, Q. Zhang, B. Du, X. Huang, Y. Y. Tang, and D. Tao, "Simultaneous spectral-spatial feature selection and extraction for hyperspectral images," *IEEE Trans. Cybern.*, vol. 48, no. 1, pp. 16–28, Jan. 2018.

[10] H. Su, B. Zhao, Q. Du, and P. Du, "Kernel collaborative representation with local correlation features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1230–1241, Feb. 2019.

[11] Y. Li, J. Qu, W. Dong, and Y. Zheng, "Hyperspectral pansharpening via improved PCA approach and optimal weighted fusion strategy," *Neurocomputing*, vol. 315, pp. 371–380, Nov. 2018.

[12] W. Li, L. Zhang, L. Zhang, and B. Du, "GPU parallel implementation of isometric mapping for hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1532–1536, Sep. 2017.

[13] Z. Wang and B. He, "Locality perserving projections algorithm for hyperspectral image dimensionality reduction," in *Proc. 19th Int. Conf. Geoinform.*, Jun. 2011, pp. 1–4.

[14] X. He, D. Cai, S. Yan, and H.-J. Zhang, "Neighborhood preserving embedding," in *Proc. 10th IEEE Int. Conf. Comput. Vis.*, Oct. 2005, pp. 1208–1213.

[15] H. R. Shahdoosti and F. Mirzapour, "Spectral–spatial feature extraction using orthogonal linear discriminant analysis for classification of hyperspectral data," *Eur. J. Remote Sens.*, vol. 50, no. 1, pp. 111–124, Jan. 2017.

[16] Z. Feng, S. Yang, S. Wang, and L. Jiao, "Discriminative spectral–spatial margin-based semisupervised dimensionality reduction of hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 2, pp. 224–228, Feb. 2015.

[17] Z. F. Shao and L. Zhang, "Sparse dimensionality reduction of hyperspectral image based on semi-supervised local Fisher discriminant analysis," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 31, pp. 122–139, Sep. 2014.

[18] F. L. Luo, H. Huang, Y. L. Duan, J. M. Liu, and Y. H. Liao, "Local geometric structure feature for dimensionality reduction of hyperspectral imagery," *Remote Sens.*, vol. 9, no. 8, p. 790, Aug. 2017.

[19] H. Pu, Z. Chen, B. Wang, and G. Jiang, "A novel spatial-spectral similarity measure for dimensionality reduction and classification of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7008–7022, Nov. 2014.

[20] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, 2015.

[21] D. Liu, S. T. Li, and L. Y. Fang, "Hyperspectral image classification by exploiting the spectral-spatial correlations in the sparse coefficients," in *Proc. 6th Chin. Conf. Pattern Recognit. (CCPR)*, vol. 483, Nov. 2014, pp. 151–158.

[22] N. Hoai Ly, Q. Du, and J. E. Fowler, "Sparse graph-based discriminant analysis for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 3872–3884, Jul. 2014.

[23] F. Zang and J. Zhang, "Discriminative learning by sparse representation for classification," *Neurocomputing*, vol. 74, nos. 12–13, pp. 2176–2183, Jun. 2011.

[24] Y. Duan, H. Huang, Z. Li, and Y. Tang, "Local manifold-based sparse discriminant learning for feature extraction of hyperspectral image," *IEEE Trans. Cybern.*, vol. 51, no. 8, pp. 4021–4034, Aug. 2021, doi: 10.1109/TCYB.2020.2977461.

[25] Z. Xue, P. Du, H. Su, and S. Zhou, "Discriminative sparse representation for hyperspectral image classification: A semi-supervised perspective," *Remote Sens.*, vol. 9, no. 4, p. 386, Apr. 2017.

[26] A. Kianisarkaleh and H. Ghassemian, "Nonparametric feature extraction for classification of hyperspectral images with limited training samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 119, pp. 64–78, Sep. 2016.

[27] J. Xu, B. Tang, H. He, and H. Man, "Semisupervised feature selection based on relevance and redundancy criteria," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 9, pp. 1974–1984, Sep. 2017.

[28] D. C. , X. F. He, and J. W. Han, "Semi-supervised discriminant analysis," in *Proc. Int. Conf. Comput. Vis.*, Oct. 2007, pp. 1–17.

[29] W. Liao, A. Pizurica, P. Scheunders, W. Philips, and Y. Pi, "Semisupervised local discriminant analysis for feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 184–198, Jan. 2013.

[30] S. L. Zhang and G. X. Yu, "Semi-supervised locality preserving projections with compactness enhancement," in *Proc. Int. Conf. Educ. Inf. Technol. (ICEIT)*, vol. 2, Sep. 2010, pp. 460–464.

[31] M. Sugiyama, T. Idé, S. Nakajima, and J. Sese, "Semi-supervised local Fisher discriminant analysis for dimensionality reduction," *Mach. Learn.*, vol. 78, nos. 1–2, pp. 35–61, Jan. 2010.

[32] Y. Duan, H. Huang, Y. Tang, Y. Li, and C. Pu, "Semisupervised manifold joint hypergraphs for dimensionality reduction of hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, early access, Jul. 24, 2020, doi: 10.1109/LGRS.2020.3009144.

[33] S. Yang, P. Jin, B. Li, L. Yang, W. Xu, and L. Jiao, "Semisupervised dual-geometric subspace projection for dimensionality reduction of hyperspectral image data," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3587–3593, Jun. 2014.

[34] W. Du, W. Qiang, M. Lv, Q. Hou, L. Zhen, and L. Jing, "Semi-supervised dimension reduction based on hypergraph embedding for hyperspectral images," *Int. J. Remote Sens.*, vol. 39, no. 6, pp. 1696–1712, Mar. 2018.

[35] F. Luo, B. Du, L. Zhang, L. Zhang, and D. Tao, "Feature learning using spatial-spectral Hypergraph discriminant analysis for hyperspectral image," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2406–2419, Jul. 2019.

[36] H. Huang, M. Chen, and Y. Duan, "Dimensionality reduction of hyperspectral image using spatial-spectral regularized sparse hypergraph embedding," *Remote Sens.*, vol. 11, no. 9, p. 1039, May 2019.

[37] F. Luo, L. Zhang, X. Zhou, T. Guo, Y. Cheng, and T. Yin, "Sparse-adaptive hypergraph discriminant analysis for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 1082–1086, Jun. 2020.

[38] S. Huang, D. Yang, Y. Ge, and X. Zhang, "Discriminant hyper-Laplacian projections and its scalable extension for dimensionality reduction," *Neurocomputing*, vol. 173, pp. 145–153, Jan. 2016.

[39] Y. Gan, F. Luo, J. Liu, B. Lei, T. Zhang, and K. Liu, "Feature extraction based multi-structure manifold embedding for hyperspectral remote sensing image classification," *IEEE Access*, vol. 5, pp. 25069–25080, 2017.

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[40] Y. Duan, H. Huang, and Y. Tang, "Local constraint-based sparse manifold hypergraph learning for dimensionality reduction of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 613–628, Jan. 2021.

[41] E. Elhamifar and R. Vidal, "Sparse manifold clustering and embedding," in *Proc. Neural Inf. Process. Syst.*, 2011, pp. 55–63.

[42] H. Huang, F. Luo, J. Liu, and Y. Yang, "Dimensionality reduction of hyperspectral images based on sparse discriminant manifold embedding," *J. Photogramm. Remote Sens.*, vol. 106, pp. 42–54, Aug. 2015.

[43] S. Hougardy, "The Floyd–Warshall algorithm on graphs with negative cycles," *Inf. Process. Lett.*, vol. 110, nos. 8–9, pp. 279–281, Apr. 2010.



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